

Quality of engineering analysis (20 points)

Criteria

- Depth of analysis
- Use of appropriate equations and calculations
- Demonstration of knowledge of aerospace design
- Explicit application of course concepts to the analysis
- Clearly stated and justified assumptions and limitations
- Meaningful interpretation and discussion of results

Grading

- **Excellent (20):** Comprehensive and insightful analysis using advanced equations and analysis tools relevant to the course and application. Results are not just calculated but critically evaluated, including consideration of limitations and implications. The analysis is not only accurate but well-integrated into the overall design and objectives. An exemplary effort is evident in demonstrating an in-depth understanding of course materials, the chosen design, and its practical application. The communication of the analysis is exceptionally clear, precise, and well-organized.
- **Satisfactory (15):** Solid analysis using appropriate equations and tools discussed in the course. Results are calculated accurately but may lack thorough critical evaluation or consideration of limitations. The analysis aligns with the overall design and objectives, although some aspects could be better integrated. A commendable effort is made to utilize course materials, but the depth of understanding could be improved. Communication of the analysis is clear but may lack some organization or precision.
- **Marginal (10):** Basic analysis using simpler equations and tools, mostly from course materials. Results may have minor inaccuracies and are not critically evaluated. The analysis shows some alignment with the design objectives but is not well-integrated. An adequate effort is made to utilize course materials but lacks depth or breadth in understanding the chosen design and its application. Communication of the analysis is mostly clear but could benefit from better organization.
- **Insufficient (5):** Inadequate or incomplete analysis using equations and tools that are not well-suited to the problem or design. Results may contain significant errors and lack any form of critical evaluation. The analysis is poorly aligned with the overall design objectives. Little to no effort is evident in utilizing course materials or understanding the design and its application. Communication is unclear, disorganized, or incomplete.
- **Incomplete (0):** Absence of engineering analysis or the use of equations and tools that are irrelevant or incorrect for the design problem. No effort is made to evaluate results, and the analysis shows no alignment with design objectives or course materials. Communication of the analysis is missing or so poorly executed that it cannot be evaluated.

Quality of Design process (20 points)

Criteria:

- Logical design methodology
- Thoughtful process
- Clearly stated decision-making criteria for design choices
- Justification for design choices
- Discussion of alternatives
- Consideration of multidisciplinary and system-level factors
- User-centered design considerations (if applicable)

Grading

- **Excellent (20):** Comprehensive and iterative design process with well-justified choices aligned to course content and engineering requirements. Decision-making is clear, including evaluation of alternatives. The process adeptly merges theory with practical application and shows adaptability to changes.
 - **Satisfactory (15):** Solid design process with mostly justified choices tied to course material and requirements. Decision-making is mostly clear, but may lack depth in evaluating alternatives. The process integrates some theory with practical application.
 - **Marginal (10):** Basic design process with minimal justification of choices. Some alignment with course material and requirements but lacks comprehensive decision-making. Limited integration of theory and practice.
 - **Insufficient (5):** Poorly structured design process with inadequate justification for choices. Little to no alignment with course material and requirements. Decision-making and integration of theory and practice are largely missing.
 - **Incomplete (0):** Absent or disjointed design process. Fails to justify choices or align with course material and requirements. No evident decision-making or theoretical grounding.
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Quality of Report (10 points)

Criteria:

- Structure and organization
- Grammar and spelling
- Effective use of tables, figures, and references
- Meets page limit
- Formatting and layout

Grading

- **Excellent (10):** Exceptionally well-structured with flawless grammar. Makes excellent use of visual aids, charts, and illustrations. The report is clear, concise, and offers deep insights. All elements are cohesive, contributing to an easily understood and insightful narrative.
- **Satisfactory (8):** Well-structured with minor grammatical errors. Makes good use of visual aids like charts and illustrations. The report is mostly clear and offers some insights, but may lack depth in places. Elements are generally cohesive.
- **Marginal (6):** Adequately structured with noticeable grammatical errors. Utilizes some visual aids, although they may not add significant value. The report is generally clear but lacks in-depth insights. Cohesiveness between elements could be improved.
- **Below Adequate (4):** Poorly structured with multiple grammatical errors. Limited or ineffective use of visual aids. The report lacks clarity and offers few, if any, insights. Elements are disjointed, making the narrative difficult to follow.
- **Incomplete (0):** Report is either absent, or so poorly executed that it cannot be meaningfully assessed. Fails in structure, grammar, and use of visual aids. Offers no insights and lacks any form of cohesiveness.

Flight test completed with successful flight (BONUS 5 points) – Pass/fail

The total is 50 points.